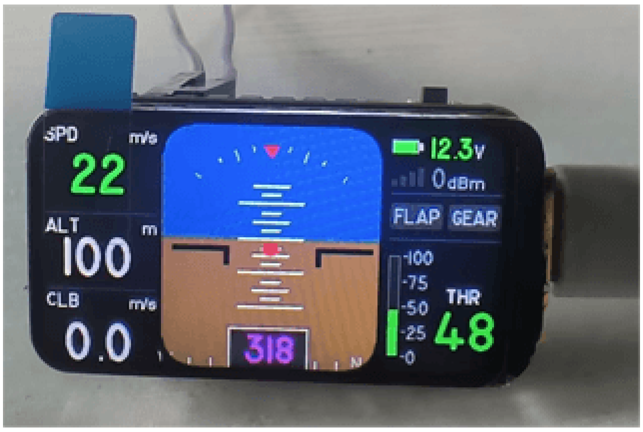
N.Y.A.F.I User Manual

Not Yet Another Flight Instrument

v1.0



- **Lightweight Electronic Flight Instrument System (EFIS)**
- **MAVLink Flight Data Display Terminal**
- **WiFi Wireless Telemetry Bridge Module**

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1. Overview

N.Y.A.F.I. (an acronym for "Not Yet Another Flight Instrument") is a lightweight Electronic Flight Instrument System (EFIS) designed to work with autopilot flight controllers running ArduPilot firmware. The device receives MAVLink v1/v2 protocol data over a serial port and displays flight parameters in real time on a color LCD.

The instrument provides four display modes: the **Data page** (DATA), the **Primary Flight Display** (PFD), the **Navigation Display** (ND) and the **Return-to-Home navigation page** (HOME). A built-in WiFi hotspot with a graphical configuration portal means no extra cabling is needed for parameter setup.

The device can also serve as a **wireless telemetry module**: it uses the built-in WiFi to bridge the flight controller's MAVLink data over UDP to a ground station (Mission Planner / QGroundControl), replacing a traditional telemetry radio. For flight controller boards without built-in WiFi or Bluetooth (such as Pixhawk, Cube, Matek and various F4/F7/H7 boards), wireless parameter tuning and data viewing only require a single TELEM serial port.

Scope of Use

- Auxiliary display for experimental aircraft and UAV ground stations
- WiFi telemetry bridging for flight controllers without wireless debugging
- External instrument panel for flight simulators
- Model aircraft / FPV telemetry data display terminal
- Education and teaching demonstration platform

> **Assembly reference**

Basic Requirements for Using This Device

- Basic model aircraft / UAV operating experience and familiarity with flight safety rules
- Ability to configure the flight controller serial port (Mission Planner / QGroundControl)
- Basic electronic wiring skills (serial, PWM, power)

> ⚠ **Disclaimer**: This device is an experimental project intended for auxiliary flight-data display only. It is **strictly forbidden** to use it as a primary flight instrument. Users assume all risks and responsibilities arising from the use of this device. The author and contributors accept **no liability** for any direct or indirect loss caused by the use of this device.
>
> ⚠ **Document vs. actual product**: The content of this document corresponds to the firmware version at the time of writing (v1.0.0). Subsequent firmware updates may cause the interface, features and parameters to differ from this description; refer to the actual device and its screen.

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2. Safety Notices

- ⚠ **Warning**: This device has NOT received any airworthiness certification. It is **strictly forbidden** to use it as a primary flight instrument — auxiliary reference only
- ⚠ **Warning**: Displayed data may be delayed, erroneous or interrupted. Make flight decisions based on the flight controller / sensor raw data
- ⚠ **Caution**: Supply voltage must not exceed 5.5V; reverse polarity may permanently damage the device
- ⚠ **Caution**: Verify the flight controller TELEM port TX/RX orientation before wiring
- ⚠ **Caution**: The WiFi hotspot has no password by default; enable encryption in sensitive environments
- ⚠ **Caution**: SD card logs contain flight data; protect privacy-sensitive information
- ⚠ **Caution**: Keep screen brightness at or below **50%** and avoid prolonged full-brightness operation. Excessive brightness raises the screen temperature and may cause dark shadows, affecting normal display
- ⚠ **Caution**: If brightness-related display anomalies occur (dark shadows, artifacts, etc.), immediately lower the brightness, let the device cool down for a while, then restart it to recover
- ⚠ **Caution**: The screen is a fragile component. Avoid impacts, crushing or forceful pressing. In vibrating environments, add protective measures as needed during installation, transport and storage (e.g. an acrylic protective plate, a cover, or foam padding)

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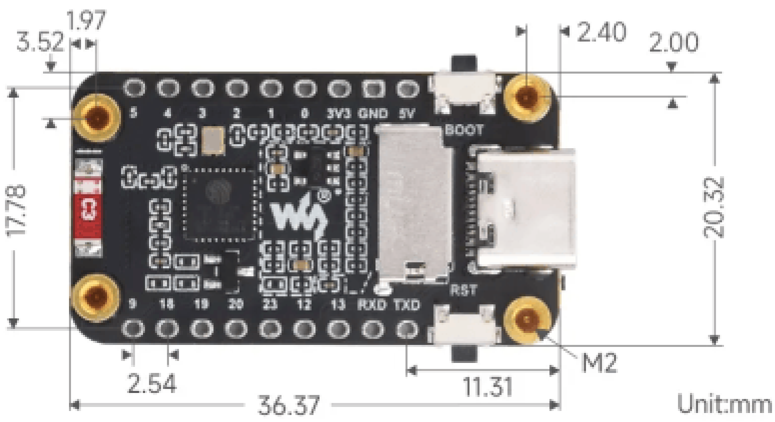
3. Interfaces

> **Product dimensions**



Board pin	Connect to	Description
RXD	Flight controller TELEM TX	Telemetry serial receive (115200 baud)
TXD	Flight controller TELEM RX	Telemetry serial transmit (optional), **required when using the telemetry function**
GPIO 23	Flap PWM signal	Two-position switch, 50 Hz, 1000–2000 μs; ≥1700μs → OPEN, <1700μs → CLOSED
GPIO 20	Landing gear PWM signal	Two-position switch, 50 Hz, 1000–2000 μs; ≥1600μs → DOWN, <1600μs → UP
GPIO 2	Page switch	3-position switch: 1000-1300μs→PFD, 1300-1700μs→ND, 1700-2000μs→HOME
GPIO 3	Backlight brightness switch	3-position switch: 1000-1300μs→25%, 1300-1700μs→50%, 1700-2000μs→80%
5V	Power input	Flight controller TELEM 5V / BEC / USB-C
GND	Ground	Must share a common ground with the flight controller

4. Installation and Wiring

Wiring Diagram

```
Flight controller TELEM      Device
┌──────────────┐        ┌─────────────┐
│  TX          ├────────→│  RXD        │
│  RX          │←────────┤  TXD        │
│  5V          ├────────→│  5V         │
│  GND         ├────────→│  GND        │
└──────────────┘        └─────────────┘
```

```
Receiver PWM              Device
┌─────────────┐      ┌────────────┐
│ Flaps       ├─────→│  GPIO 23   │
│ Gear        ├─────→│  GPIO 20   │
│ Page Sel    ├─────→│  GPIO 2    │
│ Brightness  ├─────→│  GPIO 3    │
└─────────────┘      └────────────┘
```

Wiring Notes

- The telemetry serial port connects to the board silk-screen **RXD / TXD** (the RXD/TXD pins printed on the board)
- The flight controller GND and the device GND MUST be connected, otherwise serial communication fails
- The flight controller TELEM 5V can power the device (max 500 mA); an independent BEC or USB-C may also be used
- If a PWM signal wire is longer than 30 cm, add a 100 nF filter capacitor
- Connect the flap signal wire to **GPIO 23**

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5. First Power-On

5.1 Power-On Sequence

1. Verify the wiring, then power on
2. The screen lights up showing the aircraft fly-in animation + "N.Y.A.F.I" title
3. The bottom shows the firmware version, SD card status (a prompt appears if no SD card is inserted) and communication status (a prompt appears if no telemetry is connected)
4. After ~2 seconds the animation ends and the default display page loads

5.2 Communication Status Indicator

- A flashing red **"NO COMM"** message indicates that no flight controller data has been received for over 1 second
- The message disappears automatically once communication recovers

5.3 Page Switching

Hardware method (RC switch): Connect GPIO 2 to an auxiliary channel of the receiver (3-position switch); the pulse width maps to the page. Hardware switching supports only the following **3 pages**, in the fixed order PFD → ND → HOME:

Pulse width	Page
1000-1300 μs	PFD (Primary Flight Display)
1300-1700 μs	ND (Navigation Display)
1700-2000 μs	HOME (Return-to-Home navigation)

> Note: pulse widths <800μs or >2200μs are treated as no signal and ignored.
>
> Note: The **DATA page can only be selected via the WiFi configuration portal**; the RC 3-position switch cannot reach it.

Remote method (WiFi configuration portal): All 4 pages (including DATA) can be selected via the WiFi configuration portal. The device remembers the last page used and restores it on the next power-on.

> ⚠ **Priority note**: When GPIO 2 (the RC page-switch channel) is wired and the receiver outputs a valid pulse width, the **RC channel continuously takes priority over the configuration portal**. In this case, selecting a page or setting a default page in the portal briefly shows the chosen page, then the RC channel's PWM value immediately switches it back (e.g. the switch is at center → ND; even if the portal sets PFD, the screen returns to ND right away). This is normal behavior.
>
> To make the portal-selected page effective: ① first move the 3-position switch to the position matching the desired page; ② or temporarily disconnect the GPIO 2 signal wire, configure, then reconnect it; ③ if you always use one fixed page as the default, leave GPIO 2 disconnected (then only the portal can switch pages).

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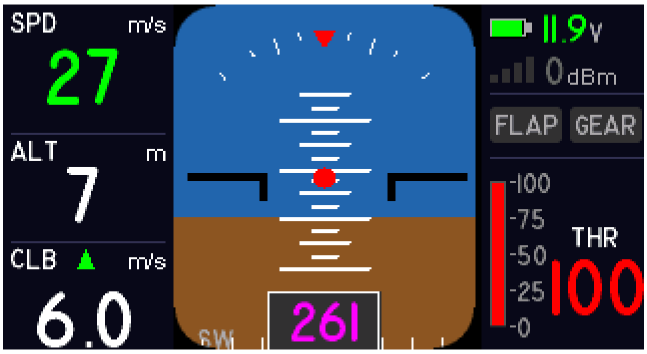
6. Display Pages

6.1 Primary Flight Display (PFD)

Page code: PFD



Left panel: Airspeed SPD (m/s, large green text), Altitude ALT (m), Climb rate CLB (m/s)

Center ADI:

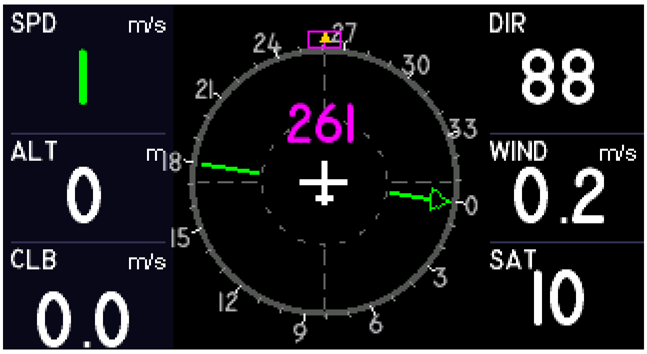
- Attitude Director Indicator, simulated gyro horizon
- Sky/ground color split background, pitch scale ticks, roll scale arc
- Heading tape at the bottom rotates with the heading

- Center aircraft symbol + red reference dot
- The red inverted triangle at the top of the roll scale arc always points down as a reference marker; the scale ticks rotate with the bank angle (±45° range)

Right panel: Battery voltage (color changes with charge level), RC signal strength RSSI, flap status, landing gear status, throttle indication (%)

6.2 Navigation Display (ND)

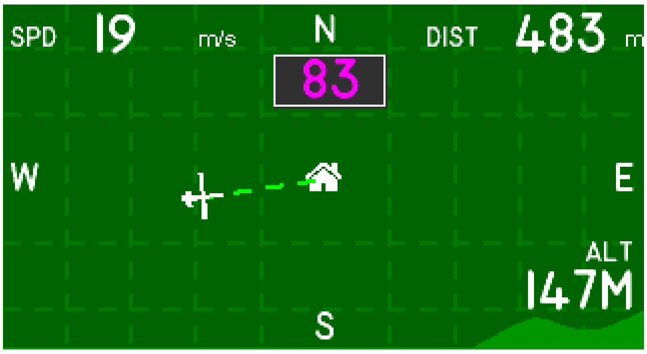
Page code: ND



- **Left panel**: Same as PFD: SPD/ALT/CLB
- **Center compass**: Pink heading indicator (ATTITUDE yaw), ticks every 5°, aircraft always points straight up, green track arrow always points toward home
- **Right panel**: Wind direction DIR (°), Wind speed WIND (m/s), Satellite count SAT ("—" when no data)

6.3 Return-to-Home Navigation (HOME)

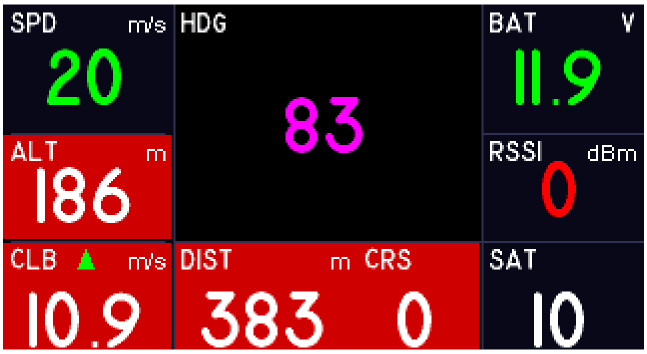
Page code: HOME



- **Top**: SPD (airspeed, m/s) and DIST (distance home, m)
- **Center**: Top-down navigation map showing the home point and the aircraft's relative position, dashed cross direction lines, auto-scaling
- **Bottom**: Altitude trend chart (60-second history) with a green filled curve; current altitude shown at the bottom right

6.4 Data Page (DATA)

Page code: DATA



Field	Meaning	Unit
SPD	Airspeed	m/s
ALT	Altitude (relative)	m
CLB	Climb rate	m/s
HDG	Heading	°
DIST	Distance home	m
CRS	Target heading	°
BAT	Battery voltage	V
RSSI	RC link signal strength	dBm
SAT	GPS satellites in view	—

The cell background flashes red when a value changes drastically.

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7. Wireless Configuration Portal

7.1 Connecting to the Hotspot

1. The WiFi hotspot starts automatically after power-on
2. Hotspot name: `N. Y. A. F. I XXXX` (XXXX is the device-unique identifier)
3. No password by default
4. After connecting, the phone/computer **pops up** the configuration page automatically, or visit `http://192.168.4.1` in a browser
5. **Hotspot timeout**: a timeout policy can be selected in the configuration page (takes effect immediately and is saved):
    - **Always on** (default): tuning mode, the hotspot stays online
    - **10 minutes**: the hotspot turns off 10 minutes after no client is connected
    - **Hotspot off**: only a 60-second configuration window at boot, then it turns off; restart the device to reopen the hotspot

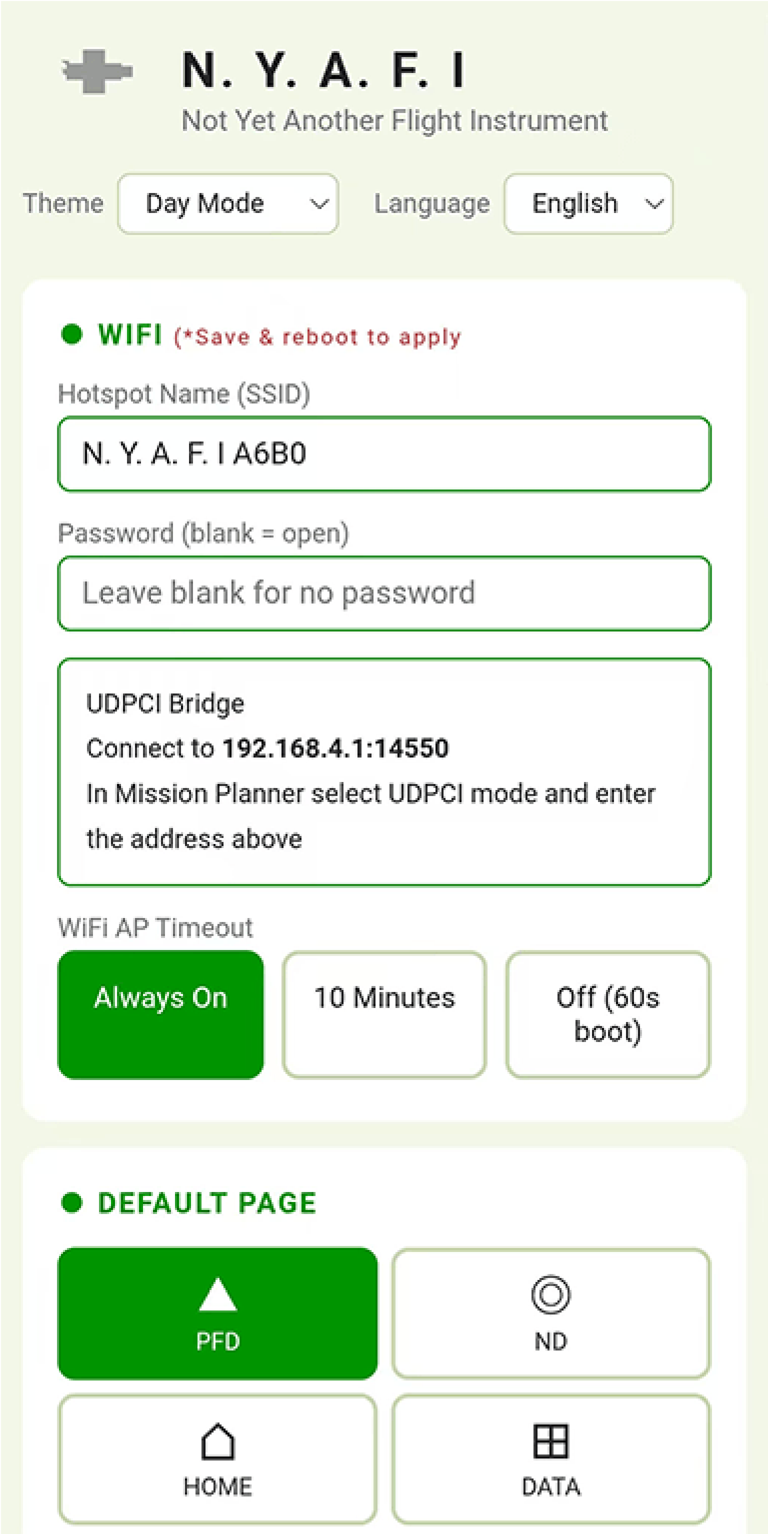
7.2 Configurable Items

Item	Description
WiFi name/password	SSID and password of the device hotspot
Default page	PFD / ND / HOME / DATA — **factory default PFD**
Language	English / 中文 / 日本語
Theme	Night mode (dark green) / Day mode (light green)
Screen rotation	0° (normal) / 180° (inverted)
Brightness level	25% / 50% / 80% — **factory default 25%**
WiFi hotspot timeout	Always on / 10 minutes / Hotspot off — **factory default Always on**

> ⚠ **Continuous use at 100% brightness increases power consumption and causes the device to heat up**; use the 25% or 50% level for daily operation. Firmware v1.0.0+ limits the maximum brightness to 80%.

The device reboots automatically after saving to apply the new parameters.

> ⚠ **If GPIO 2 is wired when setting the default page**: the RC channel's PWM value continuously decides the current page (see the priority note in 5.3); the default page saved in the configuration portal is immediately switched back. First move the RC switch to the desired page position, or temporarily disconnect the GPIO 2 signal wire before saving.



7.3 Factory Reset

- **Software method**: click "Factory Reset" at the bottom of the configuration page

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8. Ground Station Connection

The device has a built-in WiFi UDP transparent bridge listening on UDP port **14550**. After connecting to the device hotspot, the ground station can connect directly over UDP:

Ground station	Connection
Mission Planner	Select **UDPCI** → address `192.168.4.1:14550` → Connect
QGroundControl	Comm Links → Add → **UDP** → Port `14550` → OK

> UDPCI = UDP Client Initiated; MP actively connects to the device. The normal UDP mode does not work.
>
> **QGC parameter fetch fails** (after a long wait, it reports `Vehicle 1 did not respond to parameter requests`): parameter download is a single "complete set" burst transfer (the flight controller returns all several hundred `PARAM_VALUE` messages at once) and not a single one may be lost. Occasional packet loss on ordinary telemetry is unnoticeable, but if a single parameter is missing QGC keeps waiting for the full 60-second timeout. Try the following in order:
>
> 1. Set `SR1_PARAMS` ≥ 1 (recommended **10**; on ArduPilot 4.7+ search for `MAV1_PARAMS`) so the flight controller actively resends parameters;
> 2. Keep the device within 5 m of the ground-station computer with no obstructions; prefer a laptop (you can disable WiFi power saving) over a phone;
> 3. Update to the latest firmware (the serial RX buffer has been enlarged and UDP packets are limited to ≤1400 B to avoid whole-packet drops caused by IP fragmentation);
> 4. After the QGC error, click "Re-read Parameters" or reconnect.

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9. Flight Controller Setup

Configure the TELEM serial port in Mission Planner or QGC:

Parameter	Value
`SERIAL1_BAUD` (or SERIAL2)	**115** (115200 baud)
`SERIAL1_PROTOCOL`	**1** (MAVLink)

> **UI lag (horizon delay)?** An overly low telemetry stream rate from the flight controller causes display delay. Change the following parameters in MP / QGC:

Parameter	Recommended value	Effect
`SR1_EXTRA1`	**25**	ATTITUDE → **horizon roll/pitch/heading**
`SR1_EXTRA2`	**25**	VFR_HUD → airspeed/altitude/climb rate/throttle
`SR1_EXT_STAT`	**10**	SYS_STATUS (battery) + GPS_RAW_INT (satellite count)
`SR1_POSITION`	**10**	GLOBAL_POSITION_INT (GPS coordinates, ground speed)
`SR1_EXTRA3`	**10**	NAV_CONTROLLER_OUTPUT (navigation data) + HOME
`SR1_RAW_SENS`	**5**	Raw sensors
`SR1_PARAMS`	**10**	Parameter (PARAM_VALUE) stream — the key to QGC pulling the complete parameter set; factory default is usually 0 (reply-on-request only), set it to ≥1

> `SR1_*` corresponds to TELEM1, `SR2_*` corresponds to TELEM2.
>
> **ArduPilot 4.x+ note**: ArduPilot 4.0 and later renamed the `SRx_*` parameters to the `MAV_*_RATE` format (e.g. `MAV_ATTITUDE_RATE`, `MAV_VFR_HUD_RATE`, etc.). If you cannot find the `SR1_*` parameters in Mission Planner / QGC (renamed to `MAV1_*` in ArduPilot 4.7+), search for `MAV1` or `MAV2` in the parameter editor, e.g. `MAV1_EXTRA1`, `MAV1_EXTRA2`, `MAV1_POSITION`, etc.
>
> Factory defaults differ between flight controllers (Cube is usually 10 Hz, Matek F405 Wing may be only 2~4 Hz); check this item if the UI lags.

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10. Troubleshooting

Symptom	Possible cause	Solution
Screen does not light up	Insufficient power / wrong wiring	Check 5V and GND
Shows NO COMM	Serial not connected / baud mismatch / flight controller not sending data	Check the TX↔RX wiring; verify SERIALx_BAUD=115 on the flight controller; check the flight controller is powered
Values do not update	Flight controller not sending data	Check the flight controller state
Cannot connect to WiFi	Hotspot off / too far away	Restart the device; move within 5 m
Configuration page does not pop up	Device does not support Captive Portal	Manually visit `192.168.4.1` in a browser
SD card does not work	Unsupported card format	Format as FAT32; reinsert
Cannot switch pages	GPIO 2 not wired	Check the wiring; or switch remotely via the configuration portal
Default page set in the portal is switched back immediately	GPIO 2 (RC page-switch channel) is wired; the RC channel has continuous priority	Normal behavior: move the RC switch to the desired page position, or temporarily disconnect the GPIO 2 signal wire (see 5.3 / 7.2)
QGC parameter fetch timeout ("did not respond to parameter requests")	Parameter burst lost over the WiFi UDP link / flight controller parameter stream rate too low	Set `SR1_PARAMS` ≥1 (recommended 10); within 5 m using a laptop; update to the latest firmware then reconnect (see Section 8)
Values fluctuate ("float")	Sensor noise is normal when the flight controller is powered on stationary on a desk	Does not affect flight use; data stabilizes in the air; calibrate the flight controller sensors if anomalies persist in flight

> **About data fluctuation**: When the flight controller sits stationary on a desk, the barometer, GPS and other sensors still output real noise, causing ALT/CLB to jitter slightly, non-zero SPD (falls back to ground speed when airspeed=0), GPS coordinate drift, etc. **This is not a device fault**; data stabilizes substantially in flight.

Appendix

Appendix A: Pin Quick Reference

Pin	Function	Direction	Notes
RXD	Serial receive (flight controller TELEM TX)	Input	115200 8N1
TXD	Serial transmit (flight controller TELEM RX)	Output	Optional; receiving works without it
GPIO 23	Flap PWM input	Input	Two-position switch, ≥1700μs→OPEN, <1700μs→CLOSED
GPIO 20	Landing gear PWM input	Input	Two-position switch, ≥1600μs→DOWN, <1600μs→UP
GPIO 2	Page switching	Input	3-position switch pulse-width mapping (see 5.3)
GPIO 3	Backlight brightness switching	Input	3-position switch pulse-width mapping (see Section 3)
5V	Power	Input	Flight controller TELEM 5V / BEC / USB-C
GND	Ground	—	Must share a common ground with the flight controller

MAVLink serial port (flight controller communication): Serial1, default 115200, drives the MAVLink v1/v2 protocol.

End of document